Laboratorio Práctico — Packet Tracer (PDF 2)

Redes — Sección 10 · Evidencia de configuración y pruebas

1. Tablas de enrutamiento (show ip route)

R-OSPF-1

```
R-OSPF-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.1/32 is directly connected, Serial0/3/0
C       10.0.0.4/30 is directly connected, Serial0/3/1
L       10.0.0.5/32 is directly connected, Serial0/3/1
O       10.0.0.8/30 [110/128] via 10.0.0.6, 00:58:15, Serial0/3/1
O E1    10.0.0.12/30 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
O E1    10.0.0.16/30 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
O E1    10.0.0.20/30 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
O E1    172.16.0.0/24 [110/84] via 10.0.0.6, 00:05:27, Serial0/3/1
O E1    172.16.0.0/25 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
O E1    172.16.0.128/26 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
O E1    172.16.0.192/27 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
O E1 192.168.0.0/23 [110/84] via 10.0.0.6, 00:07:16, Serial0/3/1
O    192.168.0.0/24 [110/129] via 10.0.0.2, 00:58:15, Serial0/3/0
                    [110/129] via 10.0.0.6, 00:58:15, Serial0/3/1
     192.168.1.0/24 is variably subnetted, 4 subnets, 3 masks
C       192.168.1.0/26 is directly connected, FastEthernet0/0.10
L       192.168.1.1/32 is directly connected, FastEthernet0/0.10
C       192.168.1.64/28 is directly connected, FastEthernet0/0.30
L       192.168.1.65/32 is directly connected, FastEthernet0/0.30
```

R-OSPF-2

```
R-OSPF-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
C       10.0.0.0/30 is directly connected, Serial0/3/0
L       10.0.0.2/32 is directly connected, Serial0/3/0
O       10.0.0.4/30 [110/128] via 10.0.0.10, 00:58:44, Serial0/3/1
C       10.0.0.8/30 is directly connected, Serial0/3/1
L       10.0.0.9/32 is directly connected, Serial0/3/1
O E1    10.0.0.12/30 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
O E1    10.0.0.16/30 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
O E1    10.0.0.20/30 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
O E1    172.16.0.0/24 [110/84] via 10.0.0.10, 00:06:11, Serial0/3/1
O E1    172.16.0.0/25 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
O E1    172.16.0.128/26 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
O E1    172.16.0.192/27 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
O E1 192.168.0.0/23 [110/84] via 10.0.0.10, 00:08:00, Serial0/3/1
     192.168.0.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.0.0/24 is directly connected, FastEthernet0/0
L       192.168.0.1/32 is directly connected, FastEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/129] via 10.0.0.1, 00:58:44, Serial0/3/0
                       [110/129] via 10.0.0.10, 00:58:44, Serial0/3/1
O       192.168.1.64/28 [110/129] via 10.0.0.1, 00:58:44, Serial0/3/0
                        [110/129] via 10.0.0.10, 00:58:44, Serial0/3/1
```

R-CENTRAL

```
R-CENTRAL#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 10 subnets, 2 masks
O       10.0.0.0/30 [110/192] via 10.0.0.5, 00:55:06, Serial0/2/0
                    [110/192] via 10.0.0.9, 00:55:06, Serial0/2/1
C       10.0.0.4/30 is directly connected, Serial0/2/0
L       10.0.0.6/32 is directly connected, Serial0/2/0
C       10.0.0.8/30 is directly connected, Serial0/2/1
L       10.0.0.10/32 is directly connected, Serial0/2/1
D       10.0.0.12/30 [90/2048000] via 10.0.0.17, 00:55:13, Serial0/3/0
                     [90/2048000] via 10.0.0.21, 00:55:12, Serial0/3/1
C       10.0.0.16/30 is directly connected, Serial0/3/0
L       10.0.0.18/32 is directly connected, Serial0/3/0
C       10.0.0.20/30 is directly connected, Serial0/3/1
L       10.0.0.22/32 is directly connected, Serial0/3/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
S       172.16.0.0/24 is directly connected, Null0
D       172.16.0.0/25 [90/1538560] via 10.0.0.21, 00:55:12, Serial0/3/1
D       172.16.0.128/26 [90/1538560] via 10.0.0.17, 00:55:13, Serial0/3/0
D       172.16.0.192/27 [90/1538560] via 10.0.0.17, 00:55:13, Serial0/3/0
D    192.168.0.0/23 is a summary, 00:55:06, Null0
O    192.168.0.0/24 [110/65] via 10.0.0.9, 00:55:06, Serial0/2/1
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
O       192.168.1.0/26 [110/65] via 10.0.0.5, 00:55:06, Serial0/2/0
O       192.168.1.64/28 [110/65] via 10.0.0.5, 00:55:06, Serial0/2/0
```

R-EIGRP-1

```
R-EIGRP-1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/5632000] via 10.0.0.18, 00:31:31, Serial0/3/1
D EX    10.0.0.4/30 [170/5632000] via 10.0.0.18, 00:31:38, Serial0/3/1
D EX    10.0.0.8/30 [170/5632000] via 10.0.0.18, 00:31:36, Serial0/3/1
C       10.0.0.12/30 is directly connected, Serial0/3/0
L       10.0.0.13/32 is directly connected, Serial0/3/0
C       10.0.0.16/30 is directly connected, Serial0/3/1
L       10.0.0.17/32 is directly connected, Serial0/3/1
D       10.0.0.20/30 [90/5376000] via 10.0.0.18, 00:31:37, Serial0/3/1
                     [90/5632000] via 10.0.0.14, 00:31:46, Serial0/3/0
     172.16.0.0/16 is variably subnetted, 5 subnets, 4 masks
D       172.16.0.0/25 [90/1794560] via 10.0.0.14, 00:33:02, Serial0/3/0
                      [90/5378560] via 10.0.0.18, 00:31:37, Serial0/3/1
C       172.16.0.128/26 is directly connected, FastEthernet0/0.10
L       172.16.0.129/32 is directly connected, FastEthernet0/0.10
C       172.16.0.192/27 is directly connected, FastEthernet0/0.20
L       172.16.0.193/32 is directly connected, FastEthernet0/0.20
D    192.168.0.0/23 [90/5632000] via 10.0.0.18, 00:31:31, Serial0/3/1
```

R-EIGRP-2

```
R-EIGRP-2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 8 subnets, 2 masks
D EX    10.0.0.0/30 [170/5632000] via 10.0.0.22, 00:33:45, Serial0/3/1
D EX    10.0.0.4/30 [170/5632000] via 10.0.0.22, 00:33:51, Serial0/3/1
D EX    10.0.0.8/30 [170/5632000] via 10.0.0.22, 00:33:50, Serial0/3/1
C       10.0.0.12/30 is directly connected, Serial0/3/0
L       10.0.0.14/32 is directly connected, Serial0/3/0
D       10.0.0.16/30 [90/5376000] via 10.0.0.22, 00:33:51, Serial0/3/1
                     [90/5632000] via 10.0.0.13, 00:34:00, Serial0/3/0
C       10.0.0.20/30 is directly connected, Serial0/3/1
L       10.0.0.21/32 is directly connected, Serial0/3/1
     172.16.0.0/16 is variably subnetted, 4 subnets, 4 masks
C       172.16.0.0/25 is directly connected, FastEthernet0/0
L       172.16.0.1/32 is directly connected, FastEthernet0/0
D       172.16.0.128/26 [90/1794560] via 10.0.0.13, 00:35:16, Serial0/3/0
                        [90/5378560] via 10.0.0.22, 00:33:51, Serial0/3/1
D       172.16.0.192/27 [90/1794560] via 10.0.0.13, 00:35:16, Serial0/3/0
                        [90/5378560] via 10.0.0.22, 00:33:51, Serial0/3/1
D    192.168.0.0/23 [90/5632000] via 10.0.0.22, 00:33:45, Serial0/3/1
```

2. Evidencia del balanceo OSPF

R-OSPF-1: costo 128 en el enlace directo (Serial0/3/0) y 64 hacia R-CENTRAL (Serial0/3/1). Hacia 192.168.0.0/24 existen dos caminos con la misma métrica (129).

```
R-OSPF-1#show ip ospf interface

FastEthernet0/0.10 is up, line protocol is up
  Internet address is 192.168.1.1/26, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
FastEthernet0/0.30 is up, line protocol is up
  Internet address is 192.168.1.65/28, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State WAITING, Priority 1
  No designated router on this network
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    No Hellos (Passive interface)
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/3/0 is up, line protocol is up
  Internet address is 10.0.0.1/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 128
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:05
  Index 3/3, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 2.2.2.2
  Suppress hello for 0 neighbor(s)
Serial0/3/1 is up, line protocol is up
  Internet address is 10.0.0.5/30, Area 0
  Process ID 1, Router ID 1.1.1.1, Network Type POINT-TO-POINT, Cost: 64
  Transmit Delay is 1 sec, State POINT-TO-POINT,
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:07
  Index 4/4, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 1 , Adjacent neighbor count is 1
    Adjacent with neighbor 5.5.5.5
  Suppress hello for 0 neighbor(s)
```

```
R-OSPF-1#show ip route 192.168.0.0
Routing entry for 192.168.0.0/24
Known via "ospf 1", distance 110, metric 129, type intra area
  Last update from 10.0.0.2 on Serial0/3/0, 00:38:01 ago
  Routing Descriptor Blocks:
  * 10.0.0.2, from 2.2.2.2, 00:38:01 ago, via Serial0/3/0
      Route metric is 129, traffic share count is 1
    10.0.0.6, from 2.2.2.2, 00:38:01 ago, via Serial0/3/1
      Route metric is 129, traffic share count is 1
```

3. Evidencia del balanceo EIGRP

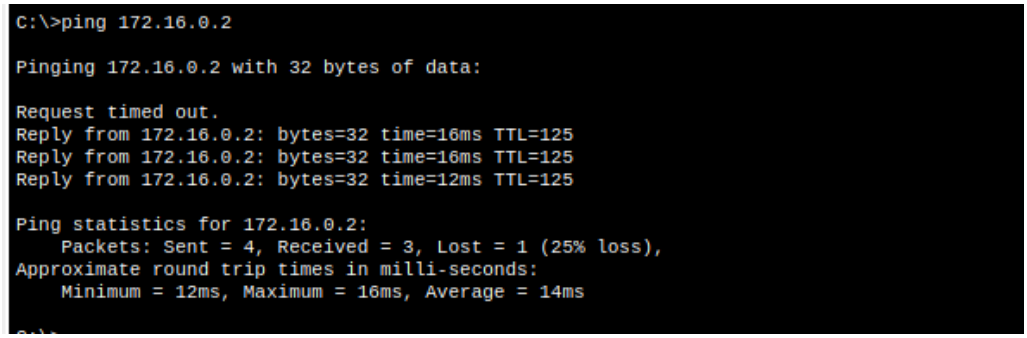
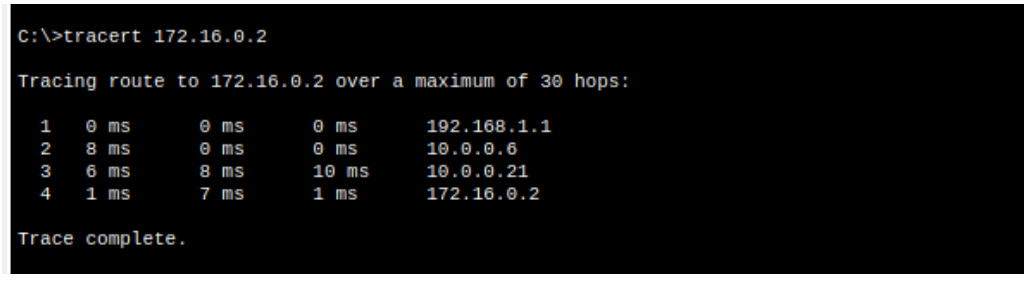
R-EIGRP-1: para 172.16.0.0/25 el successor es 10.0.0.14 (FD 1794560) y el feasible successor es 10.0.0.18 (métrica 5378560, distancia reportada 1538560 menor que la FD), instalado con variance 3.

```
R-EIGRP-1#show ip eigrp topology
IP-EIGRP Topology Table for AS 100/ID(3.3.3.3)

Codes: P - Passive, A - Active, U - Update, Q - Query, R - Reply,
       r - Reply status

P 10.0.0.0/30, 1 successors, FD is 5632000
         via Rstatic (5632000/2169856)
P 10.0.0.4/30, 1 successors, FD is 5632000
         via Rstatic (5632000/2169856)
P 10.0.0.8/30, 1 successors, FD is 5632000
         via Rstatic (5632000/2169856)
P 10.0.0.12/30, 1 successors, FD is 1792000
         via Connected, Serial0/3/0
P 10.0.0.16/30, 1 successors, FD is 5120000
         via Connected, Serial0/3/1
P 10.0.0.20/30, 2 successors, FD is 5376000
         via 10.0.0.18 (5376000/1536000), Serial0/3/1
         via 10.0.0.14 (5632000/5120000), Serial0/3/0
P 172.16.0.0/25, 2 successors, FD is 1794560
         via 10.0.0.14 (1794560/28160), Serial0/3/0
         via 10.0.0.18 (5378560/1538560), Serial0/3/1
P 172.16.0.128/26, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.10
P 172.16.0.192/27, 1 successors, FD is 28160
         via Connected, FastEthernet0/0.20
P 192.168.0.0/23, 1 successors, FD is 5632000
         via 10.0.0.18 (5632000/2169856), Serial0/3/1
```

4. Prueba de conectividad extremo a extremo

Desde PC2 (VLAN 10 de R-OSPF-1, 192.168.1.2) hacia PC5 (LAN de R-EIGRP-2, 172.16.0.2).

ping 172.16.0.2



tracert 172.16.0.2